Project Stage-1
Presentation on

"AI Research Lab in a Box: A Multi-Agent Collaboration System for Autonomous Scientific Discovery"

by

Roll Number	Group Member Names
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S013	Arya Vaidya

Under the Guidance of,

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Outline

- Introduction
- Review of the Literature
- Problem Statement
- Proposed Methodology
- Implementation and Results
- References

Introduction

- Research workflows are increasingly complex and time-consuming.
- Researchers spend significant time on repetitive tasks like filtering papers, reading methods, replicating experiments, and formatting reports.
- Existing tools handle only isolated tasks and cannot perform full autonomous research.
- There is no integrated system for both literature analysis and machine-learning experimentation.
- AI Research Lab in a Box provides a modular multi-agent framework with an NLP subsystem for literature retrieval, analysis, and synthesis.
- It also includes an ML subsystem for experiment planning, execution, evaluation, and reporting.
- The system automates the entire research lifecycle while ensuring modularity, interpretability, and scalability.

Literature Review

Sr. No	Reference Paper Title/ Existing Applications	Key Findings	Scope for improvement/Gap Identification
1	Lu, C., Lu, C., Lange, R. T., Foerster, J., Clune, J., & Ha, D. (2024). The ai scientist: Towards fully automated open-ended scientific discovery. arXiv preprint arXiv:2408.06292.	- End-to-end LLM-driven pipeline for scientific discovery. - Automates idea generation, experiment execution, paper writing, and reviewing. - Achieves near-human performance in review scoring. - Very low cost (~$15 per paper). - Applied in ML subfields: diffusion models, transformers, learning dynamics.	- Restricted to ML domain; lacks validation in medicine, biology, social sciences, and physical lab experiments. - Limited to small-scale computational experiments. - Generated references and citations may be hallucinated. - Ethical and reliability issues remain unresolved.
2	Wang, Q., Ni, S., Liu, H., Lu, S., Chen, G., Feng, X., ... & Yang, M. (2024). Autopatent: a multi-agent framework for automatic patent generation. arXiv preprint arXiv:2412.09796.	- Introduces AutoPatent, a multi-agent patent drafting system. - Draft2Patent (D2P) benchmark: 1,933 draft–patent pairs (~17k tokens). - Uses PGTree and RRAG modules for structured writing. - Qwen2.5-7B + AutoPatent outperforms GPT-4o and LLaMA-70B. - Improves drafting quality and reduces redundancy.	- Evaluation restricted to patent domain. - Still requires expert legal/technical validation. - Difficulty handling very long documents. - Limited compliance with real-world patent office standards.
3	Gridach, M., Nanavati, J., Abidine, K. Z. E., Mendes, L., & Mack, C. Agentic ai for scientific discovery: A survey of progress, challenges, and future directions. arXiv 2025. arXiv preprint arXiv:2503.08979.	- Comprehensive survey of agentic AI for research. - Reviews frameworks: LitSearch, ResearchArena, SciLitLLM, Agent Laboratory. - Applications in chemistry, biology, materials science. - Discusses evaluation metrics, datasets, implementation frameworks. - Categorizes single-agent vs. multi-agent systems.	- Literature review automation still a major challenge. - Reliability and reproducibility issues not resolved. - Limited domain-specific rigor in sensitive areas (e.g., medicine). - Ethical governance frameworks underdeveloped.

Literature Review

Sr. No	Reference Paper Title/ Existing Applications	Key Findings	Scope for improvement/Gap Identification
4	Gottweis, J., Weng, W. H., Daryin, A., Tu, T., Palepu, A., Sirkovic, P., ... & Natarajan, V. (2025). Towards an AI co-scientist. arXiv preprint arXiv:2502.18864.	- Multi-agent system (AI Co-Scientist) built on Gemini 2.0. - Uses generate–debate–evolve process with tournament ranking. - Biomedical validation: • AML drug repurposing (experimentally validated). • Novel epigenetic targets for liver fibrosis. • Rediscovery of bacterial gene transfer mechanism. - Designed for scientist-in-the-loop use.	- Dependent on heavy test-time compute scaling. - Requires wet-lab validation, slowing pipeline. - Possible risk of generating invalid but plausible hypotheses. - Not fully autonomous; expert oversight still essential.
5	Murugesan, S. (2025). The rise of agentic AI: implications, concerns, and the path forward. IEEE Intelligent Systems, 40(2), 8-14.	-Agentic AI enables autonomous, proactive, and adaptive task execution beyond traditional generative AI. -Multi-agent collaboration with role specialization can simulate complex workflows such as research or discovery. -Tool use and orchestration are central for extending LLM capabilities into structured tasks. -Applications are emerging in scientific discovery, experiment planning, and knowledge generation. -Adoption trends indicate growing interest across industries, showing timeliness for research in this area.	-Limited real-world deployment in end-to-end research workflows; most current work is fragmented. -Lack of standardized architectures for multi-agent collaboration across domains. -Explainability and accountability in agentic systems are underdeveloped, which affects trust and usability. -Few studies demonstrate robustness when agents face incomplete, noisy, or conflicting information. -Ethical and governance frameworks are still evolving, creating uncertainty for large-scale adoption.

Literature Review

Sr. No	Reference Paper Title/ Existing Applications	Key Findings	Scope for improvement/Gap Identification
6	Yang, T., Feng, P., Guo, Q., Zhang, J., Ning, J., Wang, X., & Mao, Z. (2025). AutoHMA-LLM: Efficient task coordination and execution in heterogeneous multi-agent systems using hybrid large language models. IEEE Transactions on Cognitive Communications and Networking.	-Hybrid use of large and small LLMs improves reasoning quality and efficiency for different agent roles. -Multi-agent orchestration with role specialization enhances adaptability, coordination, and scalability. -Feedback-driven task execution allows agents to self-correct, revise, and refine outputs. -Integration of reinforcement learning boosts adaptability and diversity for innovative research tasks.	-Centralized reliance on LLMs risks bottlenecks; distributed or hybrid approaches could improve robustness. -Communication overhead and high computational demands limit efficient coordination and deployment. -Scalability to larger agent networks and multi-modal input handling remain underexplored. -Ethical, safety, and real-world validation of autonomous research agents are insufficiently addressed.
7	Wang, Y., Li, T., Li, M., Cheng, Z., Wang, G., Kang, H., ... & Zhang, Y. (2025). RVBench: Role Values Benchmark for role-playing LLMs. Computers in Human Behavior: Artificial Humans, 100184.	-LLM role-playing agents often lack consistent and human-aligned values. -Role Values Dataset (RVD) captures 25 roles across 19 value dimensions from Chinese novels. -Role Values Benchmark (RVBench) evaluates agents using values rating and values ranking tasks. -Evaluation metrics show that current LLMs can simulate values but with inconsistency across dilemmas and dimensions.	-No existing benchmark comprehensively evaluates value alignment in role-playing agents. -Current LLMs struggle with consistent decision-making aligned to defined values. -More diverse datasets and cross-cultural value dimensions could improve evaluation. -Need for methods to improve LLM value alignment and stability in role-play scenarios.

Literature Review

Sr. No	Reference Paper Title/ Existing Applications	Key Findings	Scope for improvement/Gap Identification
8	Hartung, T. (2025). AI, Agentic Models and Lab Automation for Scientific Discovery–the beginning of scAInce. Frontiers in Artificial Intelligence, 8, 1649155.	- Explores transition from AI as *co-pilot* to *lab-pilot*. - Discusses multimodal agentic systems orchestrating labs + software. - Use cases: automated literature review, hypothesis generation, organoid intelligence, climate forecasting. - Highlights AI governance: EU AI Act, ISO 42001. - Frames concept of "scAInce" (AI-centered science).	- Visionary but lacks concrete implementations. - Reproducibility and lab safety concerns. - Risk of inequitable access between labs. - Limited demonstrations of real AI-controlled lab experiments.
9	D'Arcy, M., Hope, T., Birnbaum, L., & Downey, D. (2024). Marg: Multi-agent review generation for scientific papers. arXiv preprint arXiv:2401.04259.	- Introduced MARG, a multi-agent system using GPT-4 for peer-review generation. - Overcame LLM input length limitations by splitting papers across agents. - Specialized agents (clarity, experiments, impact) improved review specificity. - Reduced generic comments from 60% to 29% and produced 2.2× more helpful comments.	- High computational cost and inter-agent communication errors. - Limited to review generation—no integration with experiment design or execution. - Lack of validation in real-world peer review settings.
10	Derouiche, H., Brahmi, Z., & Mazeni, H. (2025). Agentic AI Frameworks: Architectures, Protocols, and Design Challenges. arXiv preprint arXiv:2508.10146.	- Comprehensive review of Agentic AI frameworks (CrewAI, LangGraph, AutoGen, Semantic Kernel, Agno, ADK, MetaGPT). - Compared architectures, communication protocols (CNP, A2A, ANP, Agora), memory, and safety guardrails. - Proposed taxonomy for evaluating Agentic AI systems.	- Mostly descriptive; lacks applied experiments. - No benchmarking in end-to-end scientific workflows. - Gaps in domain-specific adaptation for research automation.

Literature Review

Sr. No	Reference Paper Title/ Existing Applications	Key Findings	Scope for improvement/Gap Identification
11	Su, H., Chen, R., Tang, S., Zheng, X., Li, J., Yin, Z., ... & Dong, N. (2024). Two heads are better than one: A multi-agent system has the potential to improve scientific idea generation.	- Proposed VIRSCI, a multi-agent system mimicking teamwork in idea generation. - Workflow includes collaborator selection, topic discussion, idea generation, novelty assessment, abstract generation. - Outperformed single-agent baseline with +13.8% novelty and +44.1% research impact.	- Focus restricted to idea generation and abstract writing. - Experiments limited to text-based evaluations, no real-world validation. - Needs extension to full research cycle (methods, experiments, papers).
12	Brodimas, D., Birbas, A., Kapolos, D., & Denazis, S. (2025). Intent-Based Infrastructure and Service Orchestration Using Agentic-AI. IEEE Open Journal of the Communications Society.	-Agentic AI integrated with intent-based networking enables autonomous management, configuration, and optimization of mobile network services with minimal human intervention. -The framework successfully translates high-level natural language intents into precise network actions through a distributed multi-agent system. -Specialized agents (IDA, CIA, KRA, SMA) coordinate using tools and LLMs to provision infrastructure, manage Kubernetes clusters, and deploy services efficiently. -Experimental evaluation shows improved task completion rates, response accuracy, and operational efficiency compared to traditional manual network management.	-Limited real-world deployment testing; most evaluations are in simulated or controlled environments. -Scalability challenges may arise with larger, more complex networks or high-frequency intent requests. -Security, privacy, and fault-tolerance aspects of multi-agent coordination are not fully addressed. -Dependency on accurate intent translation and LLM performance; errors in interpretation could lead to incorrect network actions.
13	Labedzki, R. (2024). Hybrid Multi-Agent Systems as a subject of scientific research in the field of management. Available at SSRN 4985522.	- Defined Hybrid Multi-Agent Systems (HyMAS) combining human and AI agents. - Provided original definitions of intelligent agents, AI agents, and HyMAS. - Highlighted LLM-enabled natural language interfaces for human-AI collaboration. - Positioned HyMAS as a paradigm shift in management research.	- Conceptual and theoretical—no empirical testing. - Management-focused, not tailored to scientific discovery workflows. - Lack of integration with autonomous research tools.

Common Limitations

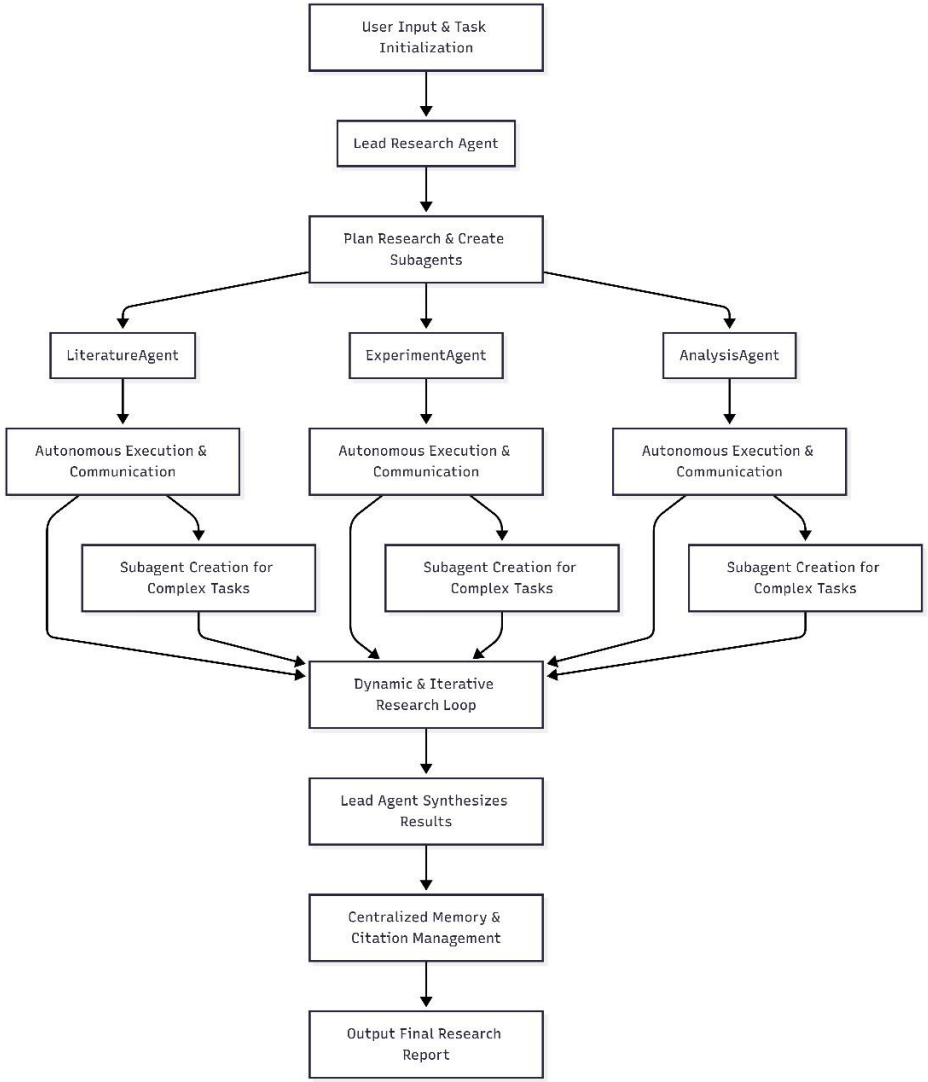
- **Domain Restriction**: Most existing systems (e.g., *AI Scientist*) are tested only in ML research or single domains (like patents, biomedical). They lack **generalization across disciplines** (medicine, social sciences, physical sciences).
- **Fragmented Research Pipelines**: Many papers handle **only one part** of research (idea generation, literature review, or peer review), but not the **entire pipeline end-to-end**.
- **Scalability Issues**: Current multi-agent systems face **communication bottlenecks, high compute cost, and inefficiency** when scaling to multiple agents or large tasks.
- **Reliability and Reproducibility**: Research outputs may be **plausible but incorrect**, citations may be hallucinated, and experimental results are not always reproducible.
- **Integration with Real Tools**: Limited connection to **external APIs (e.g., arXiv, Semantic Scholar, patent offices) or lab automation systems**, restricting practical usability.
- **Evaluation Gaps**: Few systems have **robust benchmarks** for checking novelty, coherence, and scientific soundness of outputs.
- **Human Oversight Dependence**: Most frameworks still require **scientist-in-the-loop validation**, especially in experiments and hypothesis verification, limiting true autonomy.
- **Ethics and Governance**: Lack of strong frameworks for **explainability, accountability, fairness, and governance**, making adoption risky in sensitive fields.
- **Limited Collaboration Strategies**: Role-based multi-agent coordination is underdeveloped — systems often struggle with **conflicts, redundancy, or poor division of labor** among agents.
- **Real-World Validation**: Few studies show **deployment in live research environments** (labs, industry, multi-domain research groups). Most are simulations or controlled setups.

Problem Statement

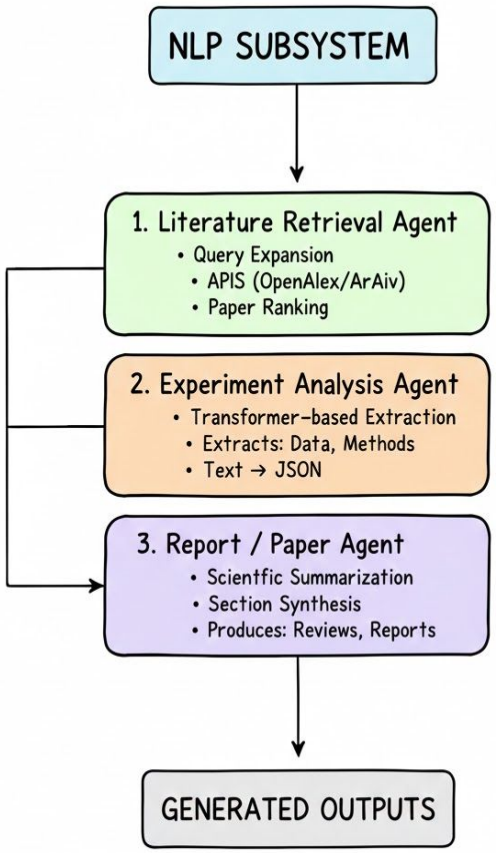
Scientific research is a multi-phase, resource-intensive process involving idea generation, literature review, experimentation, analysis, and reporting. While AI and Large Language Models (LLMs) have been applied to assist individual research tasks, the concept of fully autonomous AI agents conducting end-to-end scientific research has not been realized, leaving significant gaps in methodology, theory, evidence, empirics, and knowledge.

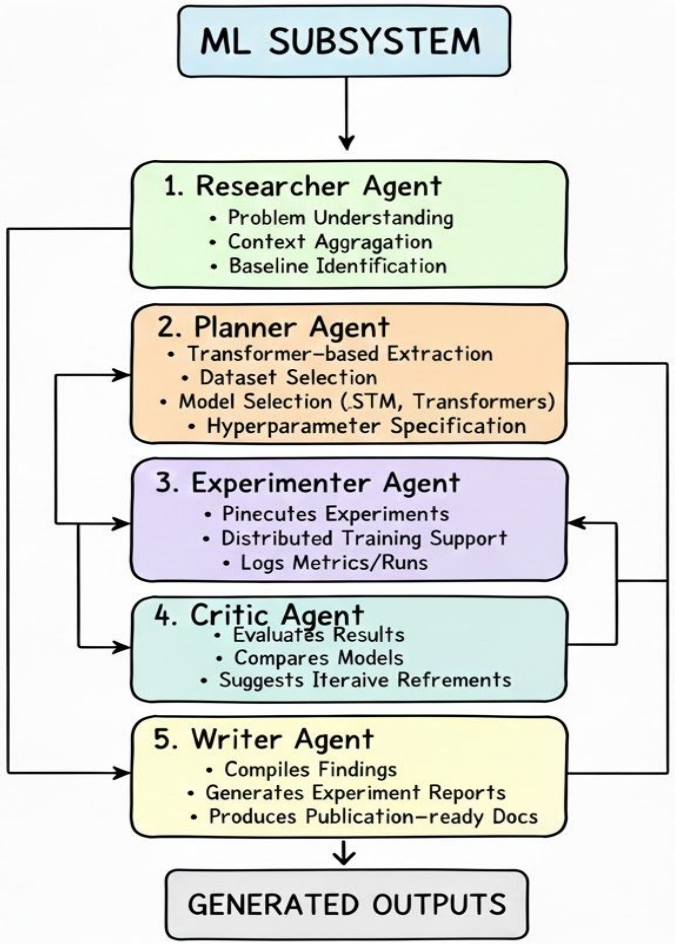
- **Methodological Gap:**
  1. Lack of frameworks for coordinating multiple AI agents in a complete research workflow.
  2. Absence of methods for autonomous task delegation and interaction among agents.
- **Theoretical Gap:**
  1. No established model explaining AI-driven scientific discovery end-to-end.
  2. Insufficient theories on inter-agent knowledge synthesis and decision-making.
- **Evidence Gap:**
  1. Limited proof of AI performing complete research processes.
  2. No comparative studies between AI multi-agent systems and human research teams.
- **Empirical Gap:**
  1. Experiments focus only on isolated tasks, not sequential or interdependent research activities.
  2. Lack of validation for scalability and reliability of autonomous research agents.
- **Knowledge Gap:**
  1. Unclear strategies for knowledge generation, validation, and integration among AI agents.
  2. Undefined protocols for role assignment and autonomous decision-making in research contexts

Proposed Methodology



Proposed Methodology





Implementations and Results

NLP Subsystem:

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```

Implementations and Results

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Implementations and Results

Report/Paper Generation Agent:

utputs > final_paper.md > # A Generalized Multi-Agent Framework for Complex Cognitive Tasks via Iterative Self-Refinement > ## 1. Introduction

A Multi-Agent Framework for Enhanced Reasoning and Reliability in Large Language Models

Abstract

Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language understanding and generation. However, they continue to face significant challenges, including factual inaccuracies (hallucinations), difficulty in complex multi-step reasoning, and susceptibility to ambiguity. This paper proposes a multi-agent system (MAS) framework designed to mitigate these limitations. By orchestrating a team of specialized AI agents, our framework decomposes complex tasks into manageable sub-problems, leveraging collaborative roles for planning, execution, validation, and refinement. We synthesize findings from recent studies that employ this paradigm across diverse applications, including autonomous solution optimization, hallucination mitigation, embodied reasoning, and systematic literature review evaluation. Experimental results from these studies demonstrate that the MAS approach significantly improves output quality, relevance, and actionability [Paper 1], shows promising outcomes in reducing hallucinations [Paper 2], and achieves high agreement with human expert evaluations [Paper 4]. This work consolidates the evidence that a structured, multi-agent architecture provides a robust pathway to more reliable and capable AI systems.

1. Introduction

The advent of Large Language Models (LLMs) based on the Transformer architecture has marked a significant milestone in artificial intelligence, achieving unprecedented success in a wide range of Natural Language Processing (NLP) tasks [Paper 3]. These models can generate coherent text, answer complex questions, and even write code. However, their widespread adoption has also illuminated several persistent limitations.

A primary concern is the phenomenon of "hallucination," where models generate confident but factually incorrect or nonsensical information, a significant challenge for their use in high-stakes applications [Paper 2]. Furthermore, ambiguity remains a fundamental challenge in NLP; LLMs can struggle to interpret and respond to user queries that are context-dependent or underspecified [Paper 5]. Finally, translating high-level, open-ended instructions into granular, actionable plans, particularly for complex or embodied tasks, remains a difficult problem [Paper 3].

To address these shortcomings, the research community is increasingly exploring Multi-Agent Systems (MAS), where multiple, specialized AI agents collaborate to achieve a common goal. This paradigm shifts from relying on a single, monolithic model to a synergistic system where different agents handle distinct aspects of a problem, such as planning, execution, critique, and refinement. This paper synthesizes evidence from a growing body of work demonstrating that such agentic frameworks can enhance the overall performance, reliability, and reasoning capabilities of AI systems. We present a consolidated framework based on these works and review experimental evidence across various domains to showcase its effectiveness.

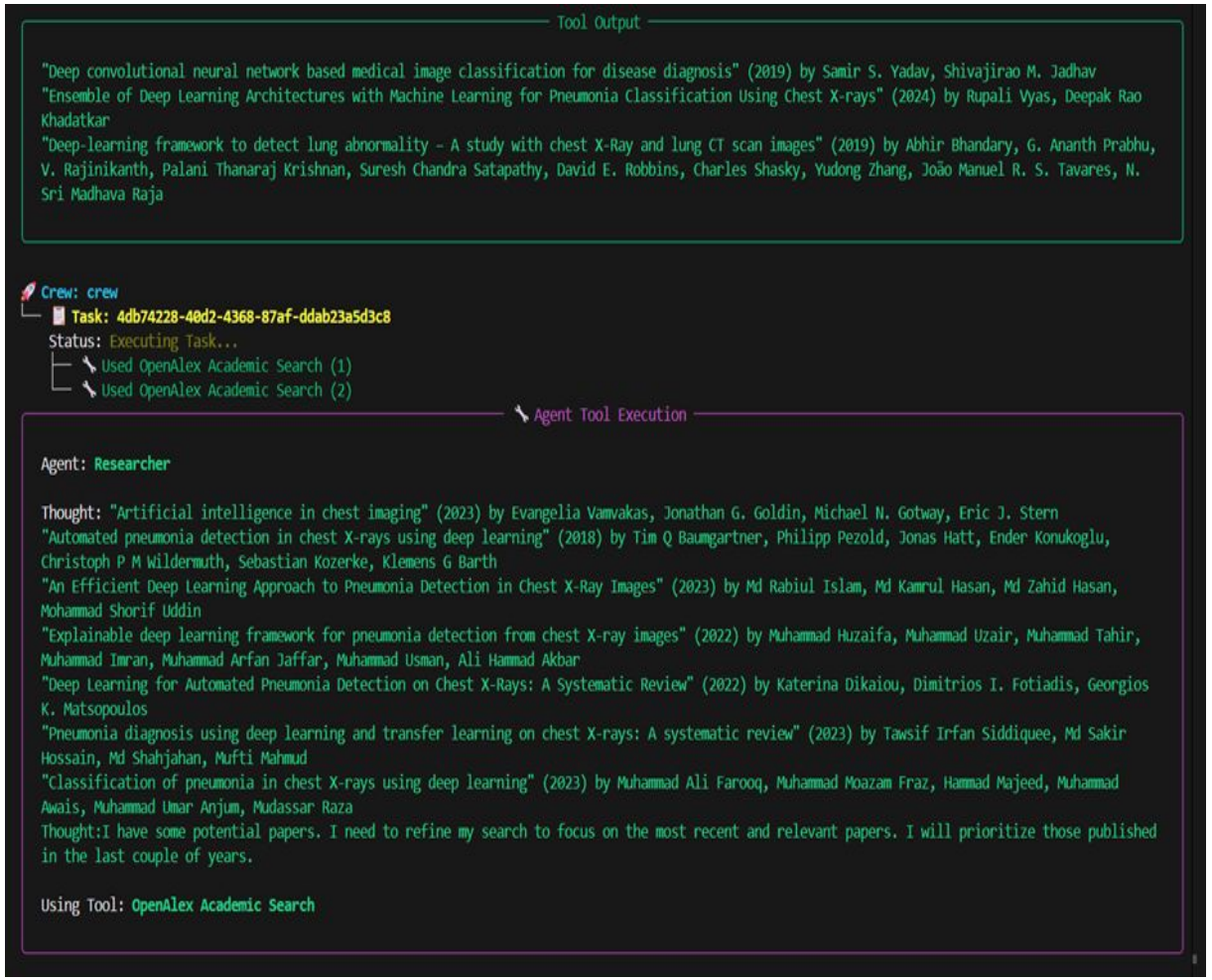
2. Related Work

The concept of using multi-agent systems to solve complex problems has gained significant traction, especially with the advent of powerful LLMs that can serve as the "brains" of individual agents. Recent literature highlights the application of this approach across several key areas.

Implementations and Results

ML Subsystem:

Researcher Agent:



Implementations and Results

Planner Agent:

Agent Final Answer

Agent: **Planner**

Final Answer:
**

Experiment Title: Ensemble Deep Learning Model with Machine Learning Integration for Enhanced Pneumonia Classification in Chest X-rays

1. Introduction

This experiment aims to develop a robust and accurate pneumonia classification model using chest X-ray images. Inspired by recent research leveraging deep learning and ensemble techniques, this experiment will combine multiple pre-trained deep learning models and integrate a machine learning classifier to further refine the classification results. The goal is to achieve state-of-the-art performance on a publicly available chest X-ray dataset.

2. Literature Review Summary

The experiment is motivated by the following research findings:

* **Ensemble of Deep Learning Architectures with Machine Learning for Pneumonia Classification Using Chest X-rays (Vyas & Khadatkar, 2024):** This paper suggests the effectiveness of combining multiple deep learning architectures in an ensemble and further enhancing the results with machine learning techniques.
* **Classification of pneumonia in chest X-rays using deep learning (Farooq et al., 2023):** This study demonstrates the potential of deep learning models for pneumonia classification.
* **An Efficient Deep Learning Approach to Pneumonia Detection in Chest X-Ray Images (Islam et al., 2023):** This work focuses on efficient deep learning models for pneumonia detection, highlighting the importance of model optimization and resource utilization.

3. Dataset

* **Dataset:** Chest X-Ray Images (Pneumonia) from Kaggle (or similar publicly available dataset). Specifically, the dataset will consist of chest X-ray images labeled as either "Pneumonia" or "Normal."
* **Data Acquisition:** Download the dataset from the specified source.
* **Data Preprocessing:**
    * **Resizing:** Resize all images to a consistent size (e.g., 224x224 pixels) to ensure compatibility with the chosen deep learning architectures.
    * **Normalization:** Normalize pixel values to a range between 0 and 1 by dividing by 255.
    * **Data Augmentation:** Apply data augmentation techniques to increase the size and diversity of the training data and improve the model's generalization ability. Augmentations may include:
        * Random rotations (e.g., -15 to 15 degrees)
        * Random horizontal and vertical flips
        * Random zoom (e.g., 0.8 to 1.2)

Implementations and Results

Experimenter Agent:

```
Task: 21dc3006-0007-401c-9da0-78eccfb7c9a6
Assigned to: Experimenter
Status: Completed
Used Multi-Architecture ML Experiment with Augmentation (1)
Agent Final Answer

Agent: Experimenter

Final Answer:
| Model          | Validation Accuracy (%) |
|----------------|-------------------------|
| resnet18 | 93.75 |
| densenet121 | 87.50 |
| efficientnet_b0 | 87.50 |

Observations:
The experiment was completed and the validation accuracies for each model are as follows: resnet18: 93.75%, densenet121: 87.50%, efficientnet_b0:
87.50%.

Task Completion

Task Completed
Name: 21dc3006-0007-401c-9da0-78eccfb7c9a6
Agent: Experimenter
Tool Args:
```

Implementations and Results

Critic Agent:

```
Agent Final Answer

Agent: Critic

Final Answer:
1. **Focus on ResNet18 and Refine Ensemble Strategy:** Prioritize the best-performing ResNet18 model. Implement a weighted averaging ensemble,
assigning a higher weight to ResNet18. Consider using ResNet18 exclusively and optimizing its hyperparameters (learning rate, optimizer, batch size)
and architecture (explore deeper ResNet variants like ResNet34/50).

2.  **Advanced Data Augmentation and Addressing Class Imbalance:** Implement advanced data augmentation techniques such as CutMix, MixUp, and
elastic transformations. Revisit the class imbalance issue using SMOTE, ADASYN, or Focal Loss to handle the potential bias towards the majority
class and improve the model's sensitivity to pneumonia cases.

Task Completion

Task Completed
Name: c9dd44a5-7c7b-4ad7-a291-b5bf552dc5fa
Agent: Critic
Tool Args:
```

Implementations and Results

Writer Agent:

Agent Final Answer

Agent: **Writer**

Final Answer:

```markdown
# Ensemble Deep Learning Model with Machine Learning Integration for Enhanced Pneumonia Classification in Chest X-rays

## 1. Introduction

This experiment aims to develop a robust and accurate pneumonia classification model using chest X-ray images. Inspired by recent research leveraging deep learning and ensemble techniques, this experiment combines multiple pre-trained deep learning models and integrates a machine learning classifier to further refine the classification results. The goal is to achieve state-of-the-art performance on a publicly available chest X-ray dataset. The initial experiment involved training and validating three deep learning models: ResNet18, DenseNet121, and EfficientNetB0.

## 2. Literature Review Summary

The experiment is motivated by the following research findings:

*   **Ensemble of Deep Learning Architectures with Machine Learning for Pneumonia Classification Using Chest X-rays (Vyas & Khadatkar, 2024):** This paper suggests the effectiveness of combining multiple deep learning architectures in an ensemble and further enhancing the results with machine learning techniques.
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*   **Data Acquisition:** The dataset was downloaded from the specified source.
*   **Data Preprocessing:**
    *   **Resizing:** All images were resized to a consistent size (224x224 pixels) to ensure compatibility with the chosen deep learning architectures.
    *   **Normalization:** Pixel values were normalized to a range between 0 and 1 by dividing by 255.
    *   **Data Augmentation:** Data augmentation techniques were applied to increase the size and diversity of the training data and improve the model's generalization ability.  Augmentations included:
        *   Random rotations (-15 to 15 degrees)
        *   Random horizontal and vertical flips
        *   Random zoom (0.8 to 1.2)
        *   Random shifts (horizontal and vertical)
```

References

1. C. Lu, C. Lu, R. T. Lange, J. Foerster, J. Clune and D. Ha, “The AI Scientist: Towards Fully Automated Open-Ended Scientific Discovery,” *arXiv preprint*, arXiv:2408.06292, 2024.
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THANK YOU